

Belief-State Structure in Reinforcement Learning for Monetary Policy

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This paper is sole-authored. It extends two LSE group projects, ST456 (deep learning) and ST455 (reinforcement learning), which I proposed and whose direction I led; the forecaster-informed belief model, the full-budget experiments, and all results reported here are my own follow-on work.

Abstract

Central banks are increasingly considering machine-learning forecasters for real-time state estimation, which raises a prior question: what happens to a learned policy rule when the policymaker observes a forecast rather than the state? We frame the central bank's problem as a partially observable Markov decision process in a three-equation New Keynesian economy, in which the agent observes a belief supplied by a forecaster-informed noise model, and we project the learned policy onto the Taylor basis. Across twelve full-budget runs, spanning two algorithms, three seeds and two calibration conditions, the long-run inflation response falls to between 0.011 and 0.025, against 1.660 for the analytical Taylor benchmark and 0.756 to 1.196 for full-observability controls. None of the twelve is statistically distinguishable from zero, with p between 0.218 and 0.518, while all twenty-four runs in which a response is present are significant at $p < 10^{-8}$. The response is therefore absent rather than merely small. We cannot distinguish the calibrated from the overconfident condition: that contrast is a comparison at the floor, and we report it as not identified. The collapse is localised to the belief model, and is consistent with attenuation under signal noise roughly five times the simulated dispersion of inflation.

Introduction: forecasts as observations

Central banks are increasingly considering machine-learning forecasters for real-time macroeconomic state estimation, and a small reinforcement-learning literature has begun to study automated policy rules. Hinterlang and Tänzer (2021) train a DDPG agent to learn an interest-rate reaction function within a neural representation of the economy estimated from data, and evaluate the result against Taylor rules; Chen et al. (2025) apply deep reinforcement learning within a monetary DSGE, though to model bounded rationality on the household side rather than to learn a central bank's rule. In both, the transition model is fixed and the learning agent conditions upon the state. Chen, Wang and Feinstein (2025) come closest to the present question, comparing tabular and function-approximation methods on a monetary-policy environment estimated from FRED data and including a partially observable variant in which a particle filter tracks a belief under Gaussian observation noise.¹

¹We cite this last as a preprint arising from a course project rather than as established work, and note that its author ordering differs between arXiv versions. It is nonetheless the nearest antecedent we have found, and the distinction drawn here is a distinction from it rather than a claim of priority over it.

What none of this work asks is what becomes of the learned rule when the belief is supplied by a *forecaster* rather than by generic observation noise, i.e. when the object upon which the policy conditions carries the error structure of a real predictive model, including a reported second moment that may or may not match the first. That is the question here, and the evaluation differs accordingly: we ask not whether the agent earns a good return but whether the rule it learns retains the determinacy property that makes a Taylor rule a Taylor rule.

It is not a trivial one, because forecaster calibration and forecaster accuracy are independent properties. Two forecasters may locate their point predictions comparably well while differing sharply in how their predictive distributions widen during regime change, react to shocks, or quantify their own ignorance. To a reinforcement-learning agent these are different observation spaces: the means may agree while the variances carry quite different information about state uncertainty. If the learned policy depends only upon the means, then calibration is irrelevant to it; if policy structure depends upon how uncertainty is communicated, then the calibration of the forecaster bears directly upon what is learned.

Our environment is the three-equation New Keynesian model (Clarida, Galí & Gertler, 1999), chosen because it supplies both an analytical benchmark policy, the Taylor rule, and a structural criterion against which a learned policy can be measured, the Taylor principle $\varphi_\pi > 1$ (Taylor, 1993). The belief the agent observes is generated by a noise model parameterised from two Bayesian deep-learning forecasters developed in LSE group coursework: a calibrated multi-scale LSTM trained upon a loss augmented with the continuous ranked probability score (Gneiting & Raftery, 2007), whose expected calibration error on inflation is 0.033, and an overconfident MC Dropout LSTM (Gal & Ghahramani, 2016) at 0.356, whose reported σ is roughly five times smaller and which contracts, rather than widens, going into the COVID-19 regime change.²

Our central result is negative, and we think the more useful for being so. Interposing a learned belief between the state and the policy destroys the determinacy property of the learned rule: the long-run inflation response collapses to roughly one per cent of the benchmark and is statistically indistinguishable from zero in every run, under both algorithms, on every seed, and in both calibration conditions. The calibration contrast we set out to measure is not identified in this design, since the two conditions differ only in the reported second moment while both suffer identical first-moment corruption, so that the comparison is made at the floor. The collapse is localised to the belief model rather than to the environment, the reward, or the algorithm, and its magnitude is consistent with attenuation under signal noise.

Setup: the environment, the belief, and the projection

The central bank sets a nominal policy rate each period in a three-equation New Keynesian economy with hybrid expectations, calibrated in annualised percentage points against a two per cent target, and is rewarded by the standard quadratic loss in the inflation gap and the output gap. Under full observability the agent observes (π_t, x_t, i_{t-1}) . Under partial observability it observes instead a belief, $(\mu_{\pi,t}, \sigma_{\pi,t}^2, \mu_{x,t}, \sigma_{x,t}^2, i_{t-1})$: a reported mean and variance for each state variable, and nothing further.

The belief loader separates two quantities that are easily conflated. The true forecast-error

²The contrast is cleanest on inflation, which is the channel the results turn upon. On the output gap the two forecasters are not reliably separated, the MC Dropout model recording the marginally lower calibration error in the pre-COVID regime.

scale τ governs the noise actually injected into the reported mean, so that $\mu_{\pi,t} = \pi_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \tau_t^2)$, whereas the reported scale σ is the uncertainty the agent is told about. Calibration, properly understood, is whether the second matches the first; both scales vary with a local volatility estimate, so that intervals widen in turbulent periods.

The design point that makes the calibration result interpretable is this: *τ is governed by identical parameters in both conditions.* The calibrated and overconfident loaders inject noise into the mean from the same distribution, with the same volatility response, and differ only in what is reported — and there in two respects, i.e. in the level of the reported σ for inflation, 2.254 against 0.456 in annualised percentage points, and in how far that reported σ tracks volatility, fully in the calibrated case and at half rate in the overconfident one.³

We evaluate by projecting the learned policy onto the Taylor basis: an OLS regression of the realised rate upon the lagged rate, the inflation gap and the output gap, with HC3 standard errors, reporting the long-run response $\varphi_\pi / (1 - \rho)$. For the partially observable conditions the regressors are the belief means the agent actually saw, since those are what the policy conditions upon. We train PPO (Schulman et al., 2017) and SAC (Haarnoja et al., 2018) for 10^6 steps under each belief condition upon three seeds, and evaluate over twenty deterministic episodes of fifty steps with the belief sequence reseeded. Alongside these we run a full-observability control, the analytical Taylor rule, and a placeholder-belief arm retained from the original coursework, in which the reported mean is near-true and carries 0.01pp of injected noise against the tier-one model’s 2.254pp.

Results: a response that cannot be detected

The metric recovers a rule where one exists. The analytical Taylor benchmark projects to a long-run inflation response of 1.660, comfortably above the determinacy threshold. Full-observability agents fall below it but remain well away from zero, at 1.196 under SAC and 0.756 under PPO. Every one of these control observations is significant at $p < 10^{-8}$. The projection is therefore capable of recovering a determinate response when the policy has one.

The collapse. Under forecaster-informed beliefs, all twelve estimates fall between 0.011 and 0.025, with condition means from 0.016 to 0.020 (Table 1). This holds under both algorithms, on all three seeds, and in both calibration conditions. Nothing approaches the determinacy threshold; the largest estimate is roughly one and a half per cent of the benchmark. The collapse is not a training failure in the ordinary sense. Every one of the twelve runs completed evaluation with zero diverged steps, and mean inflation stays within 0.16pp of target in all of them; these are stable policies that hold the economy near its steady state. What they do not do is respond to inflation. The projection’s fit is correspondingly poor, with R^2 between 0.105 and 0.417 against 0.797 to 0.843 for the analytical benchmark.

Absence, not smallness. None of the twelve is statistically distinguishable from zero, with p ranging from 0.218 to 0.518. Against this, all twenty-four runs in which a response is present, the twelve controls and the twelve placeholder-belief runs, are significant at $p < 10^{-8}$. The claim is not that the inflation response becomes small. It is that it cannot be detected at all.

Calibration is not identified. Mean long-run responses are 0.018 against 0.019 under PPO and 0.020 against 0.016 under SAC, with the sign of the difference reversing between algorithms. On

³These magnitudes are the parameters with which the belief model was instantiated, recovered from the forecasters’ saved predictions. The calibration channel in this experiment therefore runs entirely through the second moment, which is what makes a null upon it a scoped result rather than a weak one.

Table 1: Long-run inflation response φ_π by condition and seed, full-budget runs (10^6 steps, three seeds). p is the two-sided p -value on the contemporaneous inflation coefficient of the policy projection. The Taylor principle requires $\varphi_\pi > 1$.

Condition	Seed 7	Seed 42	Seed 123	Mean	p -range
Taylor rule benchmark	1.676	1.681	1.623	1.660	$< 10^{-8}$
Full observability, PPO	0.710	1.035	0.524	0.756	$< 10^{-8}$
Full observability, SAC	1.130	1.363	1.096	1.196	$< 10^{-8}$
Calibrated, PPO	0.017	0.023	0.013	0.018	0.25–0.47
Overconfident, PPO	0.023	0.023	0.011	0.019	0.28–0.52
Calibrated, SAC	0.023	0.025	0.011	0.020	0.22–0.42
Overconfident, SAC	0.018	0.020	0.011	0.016	0.33–0.49

Table 2: Mean episode reward (central-bank loss; less negative is better), partially observable conditions, full-budget runs.

Condition	Seed 7	Seed 42	Seed 123	Mean
Calibrated, PPO	-19.03	-11.70	-13.81	-14.85
Overconfident, PPO	-23.74	-13.40	-11.69	-16.28
Calibrated, SAC	-20.08	-14.62	-12.98	-15.89
Overconfident, SAC	-19.48	-15.07	-14.08	-16.21

welfare the calibrated condition holds the better mean under both algorithms and two of three seeds under each, but the margins sit inside seed-to-seed variation (Table 2). We report this as a null. A difference of 0.018 against 0.019 is a comparison between two quantities that are each indistinguishable from zero, and we draw no directional conclusion from it.

The placeholder contrast. Under the near-true-mean belief, holding the environment, the reward, the algorithms and the seeds fixed, the long-run response runs from 0.889 to 1.419 and every one of the twelve runs is significant at $p < 10^{-17}$. Because the belief model is the only thing that varies between these arms, this demonstrates that the collapse is caused by the belief representation rather than by the environment, the reward specification, or the learning algorithm. That it is caused specifically by first-moment noise, rather than by some other feature distinguishing the two loaders, is consistent with the evidence but not established by it: we did not run the arm that varies τ while holding the rest of the belief model fixed.

Interpretation: attenuation under signal noise

The magnitude of the collapse is what one would expect of a policy responding to a very noisy signal. The tier-one belief model injects noise of scale 2.254pp into the reported inflation mean, against a simulated inflation dispersion of roughly 0.43pp under the benchmark policy, a ratio near 5.3; the placeholder arm, by contrast, injects 0.01pp. A naive signal-extraction calculation, in which the response to a noisy signal is shrunk by $\text{Var}(\pi) / (\text{Var}(\pi) + \tau^2)$, gives an attenuation factor of 0.035 and predicts a long-run response near 0.058. The observed values, 0.011 to 0.025, are of the same order and somewhat below it. However, we offer this as a consistency check rather than as a test: it is a static calculation applied to a dynamic control problem, and it fixes an order of

magnitude rather than a coefficient.

The same argument explains why the calibration contrast cannot be identified here. Because τ is governed by identical parameters in both conditions, both agents receive means corrupted at the same scale, and calibration acts only upon the reported second moment, downstream of a first-moment signal the agent has already learned to discount almost entirely. A channel operating upon the reported variance has little left to act upon once the mean has been discounted to around one per cent of its benchmark weight.

The estimated dynamics corroborate this reading. The lagged-rate coefficient ρ is negative in all twelve runs under forecaster-informed beliefs, from -0.11 to -0.41 , against $+0.51$ for the Taylor benchmark and positive values in all twelve placeholder-belief runs. The learned rate oscillates rather than smooths, i.e. the opposite of the interest-rate smoothing found in estimated policy rules, and the sign flips with the belief model.

Limitations: what the design cannot settle

Three seeds is a small sample, and we perform no formal inference across conditions; the per-seed values of Table 1 are reported directly, without bootstrap intervals or pairwise tests, and the calibration comparison should be read as a null rather than as an established equivalence.

The noise scale is the sharpest constraint upon external validity. The value of 2.254pp is a forecast-error scale measured upon real US inflation data, imported into a synthetic economy whose inflation dispersion under the benchmark policy is roughly 0.43pp, so that the ratio near five is a property of that pairing rather than a claim about the signal-to-noise environment any real central bank faces. Our result should be read as conditional upon it: where belief noise is several times the dispersion of the state being estimated, the learned rule loses its inflation response. Relatedly, τ is a design parameter rather than a quantity estimated from a live forecaster; the belief model is forecaster-informed, not the forecaster itself.

The environment is a linearised benchmark that abstracts from nonlinearity, the zero lower bound, and fiscal-monetary interaction. The full-observability control is itself imperfect, since under PPO on seed 123 the policy destabilised, diverging for 17 evaluation steps with a mean episode reward of -56.52 .

Finally, we restrict the structural claim to the inflation response. The output-gap coefficient is wrong-signed in the partially observable runs; however, it is also wrong-signed in two of three full-observability PPO runs, and we therefore do not claim that the agent abandons the Taylor structure as a whole.

References

- Chen, M., Cont, R., Joseph, A., Kumhof, M., Pan, X., Xiong, W., & Zhou, X. (2025). Deep reinforcement learning in a monetary model. *Bank of England Staff Working Paper* No. 1142.
- Chen, S., Wang, T., & Feinstein, K. (2025). Reinforcement learning for monetary policy under macroeconomic uncertainty: analyzing tabular and function approximation methods. arXiv:2512.17929.
- Clarida, R., Gali, J., & Gertler, M. (1999). The science of monetary policy: a New Keynesian perspective. *Journal of Economic Literature*, 37(4), 1661–1707.
- Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning* (pp. 1050–1059).
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- Haarnoja, T., Zhou, A., Abbeel, P., & Levine, S. (2018). Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning* (pp. 1861–1870).
- Hinterlang, N., & Tänzer, A. (2021). Optimal monetary policy using reinforcement learning. *Deutsche Bundesbank Discussion Paper* No. 51/2021.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.
- Taylor, J. B. (1993). Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39, 195–214.

Note on tools

The environment, the belief model, the experiments and all analytical and directional decisions reported here are the author’s own. Claude (Anthropic) was used to verify every reported figure against the repository artifacts, to identify errors in an earlier draft of this excerpt, and to draft prose from the author’s section-level outline; the citations above were checked against primary sources following that drafting, and one erroneous pair of references was corrected as a result. The author reviewed and revised all text and is responsible for its content.